

Object-Oriented Programming

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Workshop No. 3 — Object-Oriented SOLID Principles

Universidad Distrital Francisco José de Caldas

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1 Project Definition

1.1 Project Name

Prestige Eventos & Recepciones

1.2 Description

This project is for people planning a big party, like a wedding or a large birthday. Right now, planning something like this is hard because you have to call many different people (for the place, the food, and the decorations) and try to put all the pieces together yourself. This system makes it easy. It's a single website where you can see what dates are available, choose a party theme, pick the food you want, and add other services. You get to see the total price right away and book everything at once. The main goal is to take the stress out of planning a big event, making it a simple and clear process for everyone.

1.3 Main Features

The system guides you through planning your entire event in one place. You can start by choosing a fun theme for your party, like a classic wedding or a tropical birthday, which suggests matching decorations. Then, you'll pick the perfect food and drinks for your guests, from a formal dinner to a casual buffet. Finally, you can check the availability of your favorite venue, lock in your ideal date, and tell us how many people are coming to get everything officially booked. To make it easy, you can create a personal account to save your ideas and manage all your party details securely.

2 Requirements

2.1 Functional Requirements

- User registration and authentication: Customers can create an account and log in to manage their reservations.
- Date calendar: The user can check availability on an interactive calendar.
- Event booking: The client selects the date, type of party, number of guests, and additional services.(now **Reservation** coordinates with abstractions **IBookable** and **IPriced**)
- Modify: The user must be able to cancel or modify their reservation within two weeks.
- Support: The user should be able to communicate with us in the way that best suits them.
- Service Packages: The system must allow users to view and select predefined service packages that combine venue, theme, and catering options.
- Online Payments: The user must be able to securely complete online payments for their reservations and receive immediate payment confirmation.

- **Automatic Notifications:** The system must automatically generate and send email confirmations for both completed bookings and any subsequent modifications made to the reservation.

2.2 Non-Functional Requirements

- **Ease of Use:** The design must be simple and clear so that anyone, even without experience, can easily book their event.
- **Mobile-Friendly:** The website must work perfectly and look good on any device, like smartphones and tablets.
- **Security:** User data and payment information must be completely protected with modern security measures.
- **Scalability:** The system must work smoothly even when many people are using it at the same time without slowing down or crashing.
- **Security:** User data and payment information must be completely protected with modern security measures.
- **Availability:** The system must be working and available to users 24/7, especially on weekends when most people plan events.

3 User Stories

User Story 1

Title	Choose the Best Party Package
As a	client,
I want to	see and compare different pre-defined party packages (that include a theme, food, and decorations),
so that	I can easily choose the one that best fits my needs and vision without having to plan every detail myself.
Priority	High
Effort	5
Acceptance Criteria	• The main page must show a gallery of available party packages. • Each package must display a clear title, a main image, a short description, and a starting price. • I must be able to click on a package to see a detailed page with a full list of what's included (venue, food options, decoration style, etc.).

User Story 2

Title	Check Availability on a Calendar
As a	client,
I want to	see an interactive calendar that shows which dates are available for my chosen venue,
so that	I can quickly find and select a date that works for me and my guests.
Priority	High
Effort	3
Acceptance Criteria	• The calendar must be visually clear, with available dates marked differently from booked dates. • Clicking on an available date should allow me to proceed with the booking for that date. • If a date is unavailable, it should be clearly marked and not selectable.

User Story 3

Title	Modify or Cancel a Reservation
As a	customer,
I want to	cancel or make changes to my reservation (like the number of guests or food menu),
so that	I have the flexibility to adjust my plans if my needs change.
Priority	Medium
Effort	5
Acceptance Criteria	• From 'My Account,' I must see a list of my active reservations with an 'Edit' and 'Cancel' button. • I can only cancel or change my reservation up to 14 days before the event date. • After any change, the system must show me an updated summary and a new total price. • I must receive an email confirmation after any modification or cancellation.

User Story 4

Title	Create a Personal Account
As a	customer,
I want to	create a personal account using my email and a password,
so that	I can save my preferences, see my booking history, and make future reservations faster.
Priority	High
Effort	3
Acceptance Criteria	• The registration form must only ask for my full name, email, and a secure password. • I must receive a verification email to activate my account. • Once logged in, I must have a "My Account" page where I can see my profile and past bookings.

User Story 5

Title	Contact the Organizers Easily
As a	customer,
I want to	have multiple, clear ways to contact customer support (like a contact form and a phone number),
so that	I can get help in the way that is most convenient for me if I have a question or problem.
Priority	Low
Effort	2
Acceptance Criteria	• The website's footer and contact page must display a customer service phone number and email address. • A contact form must be available that sends my message directly to the support team. • After submitting the form, I must see an on-screen confirmation that my message was sent.

4 CRC Cards

Class: Venue	Class: User
Responsibilities: • Save place info • Check dates • Verify capacity • Manage calendar	Responsibilities: • Create accounts • Save user data • Manage preferences
Collaborators: • Reservation • Calendar	Collaborators: • Reservation • Payment

Class: Calendar	Class: Payment
Responsibilities: • Show dates • Block dates • Date selection	Responsibilities: • Process payments • Payment methods • Generate receipts • Handle refunds
Collaborators: • Venue • Reservation	Collaborators: • Reservation • User

Class: Service	Class: Reservation
Responsibilities: • Manage services • Save prices • Check availability • Calculate cost	Responsibilities: • Create reservations • Calculate total • Change status • Manage dates
Collaborators: • Reservation	Collaborators: • User • Venue • Service • Payment

Interface: IPayment	
Responsibilities: • Define a standard method to process payments abstractly. • Allow multiple implementations for different payment methods.	Collaborators: • Reservation

Interface: INotification	
Responsibilities: • Define an abstraction for sending notifications to users. • Ensure flexible communication mechanisms (email, SMS, etc.).	Collaborators: • User • Reservation

Abstract Class: AbstractService	
Responsibilities: • Define common attributes and behaviors for all services. • Allow extensions without modifying existing service logic.	Collaborators: • Service • Reservation

Class: CateringService (extends AbstractService)	
Responsibilities: • Handle catering-specific booking and availability logic. • Provide custom pricing for catering events.	Collaborators: • Reservation • Venue

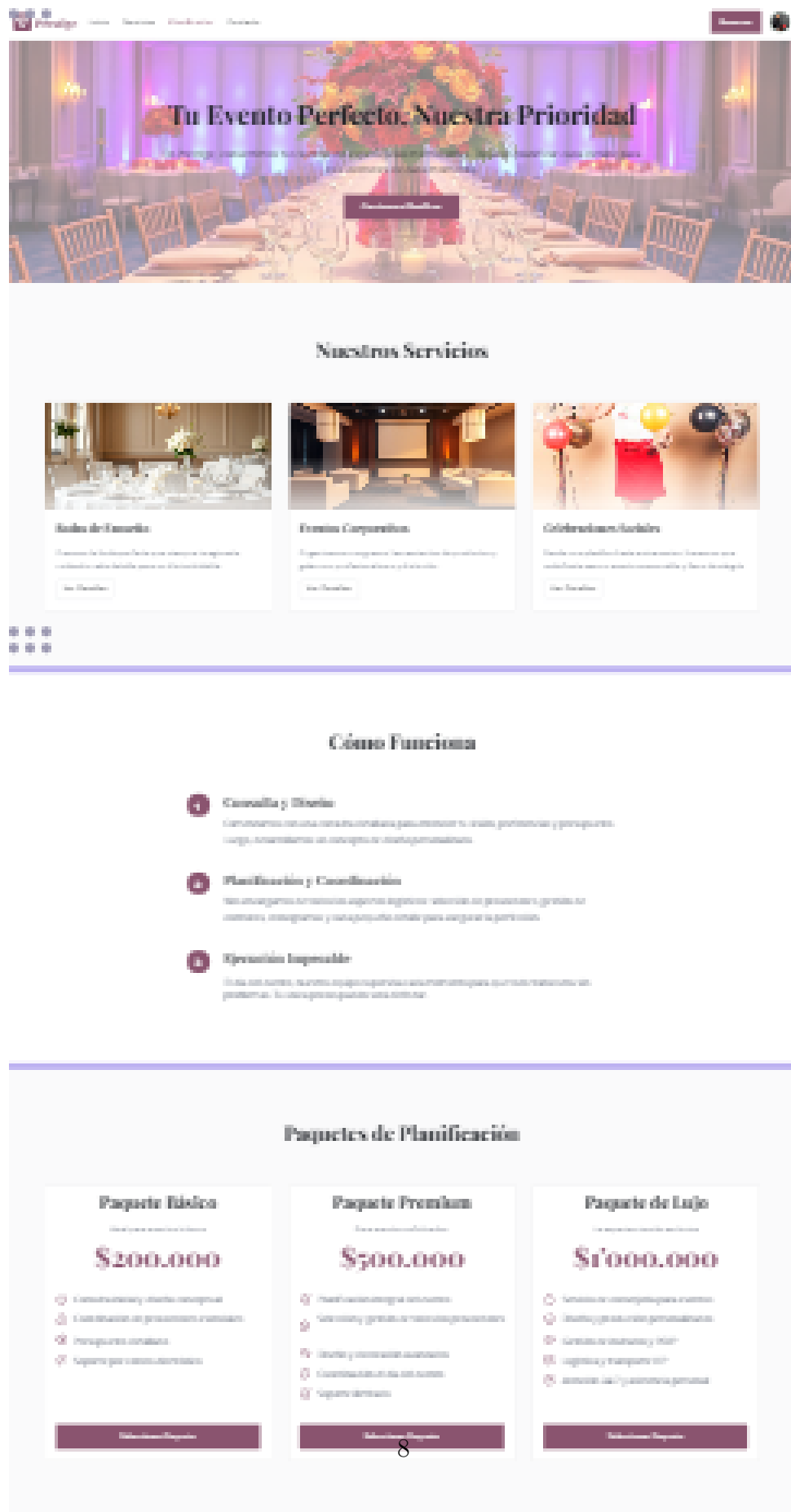
Interface: IBookable	
Responsibilities: • Define methods to verify and book availability for a service or venue.	Collaborators: • Reservation • Service

4.1 Solid -Focused Implementation

- Single Responsibility : payment no longer sends confirmation; NotificationService now sends the confirmation
- Open/close : created interfaces for new services(catering, decoration , audio)
- Liskov substitution : we introduce AbstractService and concrete Service
- interface Segregation :IBookable, IPriced, IPaymentProcessor, INotification replace thelarge interfaces.
- Dependency Inversion : high level modules depend on abstraction IPaymentProcessor, INotification

5 Mockups

User Interface Designs



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Ensalada Caprese Fresca

Una deliciosa mezcla de tomates cherry maduros, queso mozzarella fresco y hojas de albahaca, adornada con garbanzo italiano. Perfecta como acompañamiento o entrante.

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Filete Mignon con Salsa de Champiñones

Deliciosos cortes de filete de res seleccionados cocinados a la perfección, servidos con una salsa cremosa de champiñones. Acompañados de espárragos frescos y puré de papa.

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Tarta de Limón con Merengue Suave

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Paella Valenciana Tradicional

Auténtica paella valenciana con pollo, conejo, judías verdes y garbanzo, sazonada con azafrán para lograr el color y aroma perfectos. Un clásico que combina a la perfección los sabores.

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Tabla de Quesos Artesanales

Una selección asombrosa de quesos artesanales de diferentes variedades, acompañados de frutos secos, miel, mermelada y pan artesanal. Perfecta para compartir y disfrutar.

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Brochetas de Camarones al Ajillo

Yagres cremosos marinados en ajo, limón y hierbas, servidos a la parrilla hasta alcanzar una perfecta textura y sabor. Una entrada elegante y llena de sabor.

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Leche Cuajada Bolognesa

El queso de punta firme, una delicia tradicional hecha con leche de vaca y nata, y una deliciosa salsa de carne de res y tomate. Una combinación perfecta para disfrutarla sola o acompañada de pan.

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Mesa de Postres Variados

Una selección personalizada de postres que incluye: macarons, tartas de chocolate, mini cheesecakes y frutas frescas. Ideal para cualquier evento especial.

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Bebidas Refrescantes Naturales

Una selección variada de jugos naturales, de sabores clásicos y agua mineral de temperatura. Perfectos para acompañar cualquier comida y mantenerse hidratados.

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¿Listo para planificar tu evento?

Contáctanos hoy para diseñar un menú perfecto que se adapte a tus invitados. Nuestro equipo está listo para asesorarte en cada detalle.

[Contáctanos Ahora](#)



Contáctenos

No se encuentre la solución de usted. Escríbenos un mensaje y nos pondremos en contacto pronto.

Nombre

Correo Electrónico

Teléfono

Mensaje

Enviar Mensaje

Nuestra Ubicación

Encuentranos en el corazón de la ciudad para sus eventos.



Detalles de Contacto

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Nombre

Correo Electrónico

Teléfono

Mensaje

Enviar Mensaje

Nuestra Ubicación

Encuentranos en el corazón de la ciudad para sus eventos.





Solicita tu Cotización Personalizada

Planifica el evento de tus sueños con facilidad.
 Completa nuestro formulario para recibir una
 cotización detallada y adaptada a tus necesidades.
 Podemos ayudarte a hacer realidad tu visión.



Detalles del Evento

Completa estos campos para que podamos ayudarte.

Tipo de Evento

Numero de Invitados

Fecha del Evento

Temas y Preferencias

Elige los temas que te interesan y añade cualquier preferencia especial.

Selecciona temas favoritos:



Wedding



Birthday



Corporate



Party



Formal

Preferencias especiales:

Información de Contacto

Proporciona datos para que podamos contactarte.

Nombre completo

Correo electrónico

Numero de Teléfono

Resumen de tu Solicitud

Resumen de los datos que has introducido en tu solicitud.

Detalles del Evento

Tipo	No especificado
Invitados	No especificado
Fecha	No especificado

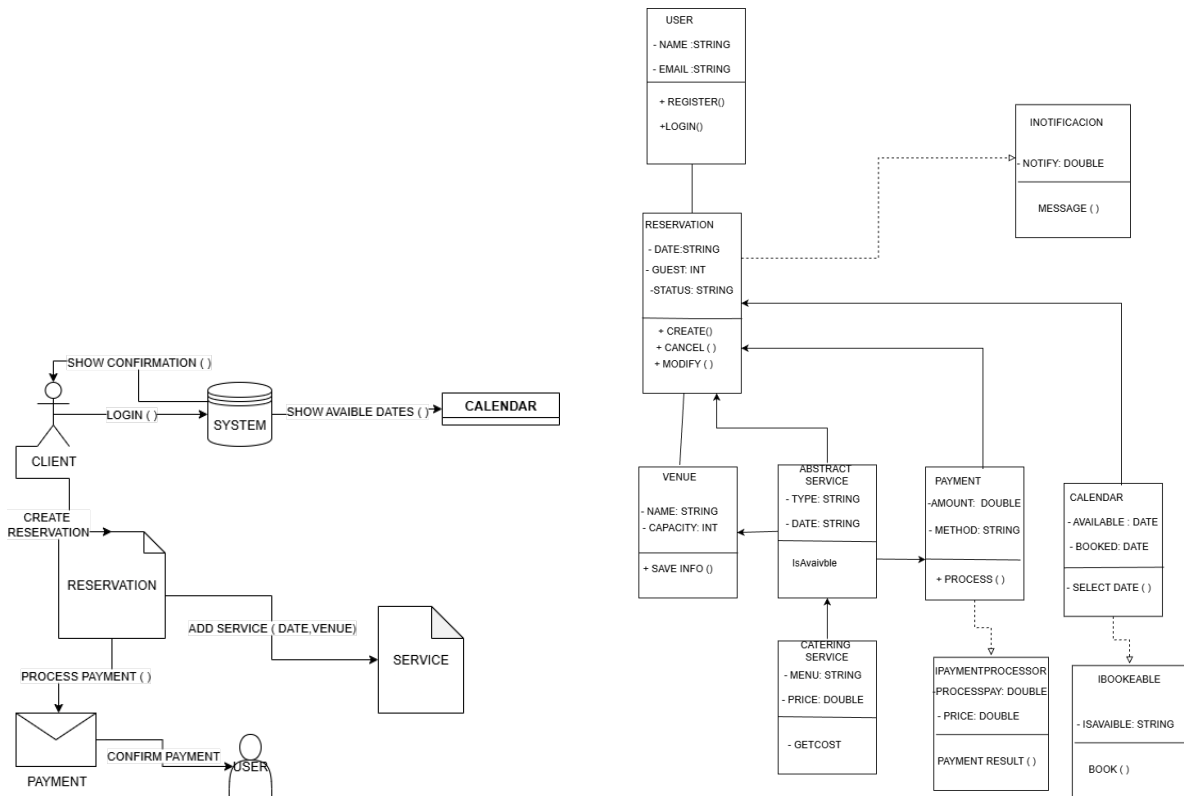
Preferencias de Tema

Temas seleccionados:	Ninguno
Preferencias especiales:	Ninguna

Datos de Contacto

Nombre	No especificado
Email	No especificado
Teléfono	No especificado

6 UML DIAGRAMS



7 Java Implementation - Applying SOLID Principles

This section show how the original system was improved using the SOLID principles, based on the existing base class. Each principle is explained, along with the changes made and the specific classes where they were applied.

7.1 Single Responsibility Principle (SRP)

Original Issue: The Payment class had two responsibilities: processing payments and sending confirmations.

Improvement: A new class called NotificationService was created to handle all message sending and confirmations. Now, Payment only focuses on payment logic.

```
// New class to apply SRP
public class NotificationService {
    public void sendConfirmation(Client client, String message) {
        System.out.println("Sending confirmation to " + client.getName() + ": " + message);
    }
}
```

```
// Modified Payment.java
public class Payment {
    private double amount;
    private boolean isProcessed;
    private NotificationService notificationService;

    public Payment(double amount, NotificationService notificationService) {
        this.amount = amount;
        this.notificationService = notificationService;
        this.isProcessed = false;
    }

    public void processPayment() {
        this.isProcessed = true;
        System.out.println("Payment of $" + amount + " processed");
    }

    public void confirmPayment(Client client) {
        if (isProcessed)
            notificationService.sendConfirmation(client, "Payment confirmed!");
        else
            System.out.println("Payment not processed yet");
    }
}

public class Client {
    private String name, email, password;
    public Client(String name, String email, String password) {
        this.name = name; this.email = email; this.password = password;
    }
    public boolean login(String email, String password) {
        return this.email.equals(email) && this.password.equals(password);
    }
    public void showConfirmation(String message) {
        System.out.println("Confirmation: " + message);
    }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
}

```

Where applied:

Responsibility for notifications was moved to a new class (NotificationService).

Payment now has a single, focused responsibility.

7.2 Open/Closed Principle (OCP)

Original Issue: The Service class was static and not easily extendable. Adding new types of services (like decoration or audio) required modifying the base code.

Improvement: Created an AbstractService class and new subclasses (e.g., CateringService) that extend it. Now, new services can be added without changing the base class.

```
//Abstract base class
public abstract class AbstractService {
    protected String name;
    protected double price;

    public AbstractService(String name, double price) {
        this.name = name;
        this.price = price;
    }

    public abstract void addService(String date, String venue);
}

//Example subclass applying OCP
public class CateringService extends AbstractService {
    public CateringService(double price) {
        super("Catering", price);
    }

    @Override
    public void addService(String date, String venue) {
        System.out.println("Catering service booked for " + date + " at " + venue);
    }
}
```

Where applied:

The old Service class was replaced with an extensible hierarchy.

Future services can be added by subclassing, not by editing existing files.

7.3 Liskov Sustitution Principle (LSP)

Implementation: Because all services now inherit from AbstractService, any subclass can be used interchangeably where the base type is expected.

```
//Example in Main.java
AbstractService service = new CateringService(500.0);
service.addService("2025-11-20", "Main Hall");
```

Where applied:

CateringService can replace AbstractService without issues.

This maintains consistent behavior and supports polymorphism.

7.4 Interface Segregation Principle (ISP)

Original Issue: There were no interfaces; classes were highly coupled and dependent on each other.

Improvement: Created smaller, more specific interfaces for different functionalities.

```
//New small interfaces
public interface IBookable {
    boolean checkAvailability(String date);
    void book(String date);
}

public interface IPriced {
    double calculatePrice();
}

public interface IPaymentProcessor {
    void process(double amount);
}

public interface INotification {
    void send(String message);
}
```

Where applied:

Each class now implements only the interfaces it needs.

Prevents unnecessary dependencies and improves modularity.

7.5 Dependency Inversion Principle (DIP)

Original Issue: The Main class directly instantiated concrete classes like Payment and Calendar, creating tight coupling. **Improvement:** Created interfaces for key behaviors (IPaymentProcessor, INotification) and updated classes to depend on these abstractions instead of specific implementations.

```
// Implementation example for DIP
public class OnlinePaymentProcessor implements IPaymentProcessor {
    @Override
    public void process(double amount) {
        System.out.println("Processing online payment of $" + amount);
    }
}

// New class depending on abstraction
public class PaymentService {
    private IPaymentProcessor processor;

    public PaymentService(IPaymentProcessor processor) {
        this.processor = processor;
    }
}
```

```

    }

    public void executePayment(double amount) {
        processor.process(amount);
    }
}

```

Where applied:

PaymentService depends on the abstraction IPaymentProcessor.

Enables easy replacement of payment methods without changing the high-level logic.

7.6 REFLECTION

we do some changes with solid structure to improve the code, the organization, the visuals and reuse of the code, also we do some changes in UML diagrams and crc cards showing these changes, we are trying to improve our app with all seeing in class and using references of other projects and pages already existing for example **eventos bella suiza,cocteles y eventos,and comun y silvestre**.